

From atomic entities to physico-chemical systems

Study guide

Concepts in detail · Recaps · Worksheets

Semester 1 · Academic Year 2026–2027

What is in this guide

- Five units, one per chapter of the student handout (pages 1–20). Each unit has three parts:
 - **Concepts in detail:** every notion of the chapter with a precise definition, the reasoning behind it, worked examples and common mistakes. **Orange boxes** complete the blank boxes and tables of the handout.
 - **Recap:** one page with the definitions, formulas, key numbers and a checklist.
 - **Worksheet:** 15 questions that get progressively harder.
- Worked answers to every question are in a separate booklet, *Chem-S1-1_Worksheet-Answers.pdf*. Try the questions first.

Contents

Unit	Title	Handout
1	Classical description of the atom	ch. 1 (1.1–1.3)
2	Quantum model of the atom	ch. 2 (2.1–2.5)
3	Quantum numbers and atomic orbitals	ch. 3 (3.1–3.4)
4	Electronic configuration and periodic table	ch. 4 (4.1–4.2), pp. 13–15
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Worksheet levels

Level 1 ●	Q1–4: recall a definition, apply one formula.
Level 2 ●●	Q5–8: explain a result or chain two steps.
Level 3 ●●●	Q9–12: multi-step problems, interpreting data.
Challenge ★	Q13–15: synthesis, links between units, going beyond the handout.

Constants as in the handout: $c = 3.00 \times 10^8 \text{ ms}^{-1}$, $h = 6.62 \times 10^{-34} \text{ Js}$, $e = 1.6 \times 10^{-19} \text{ C}$, $R_H = 3.29 \times 10^{15} \text{ Hz}$, $a_0 = 52.9 \text{ pm}$, $1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$. Where the handout has a typo (for example the Balmer formula, or “650” written for 700 nm on the spectra), the guide says so.

UNIT 1

Classical description of the atom

Handout chapter 1, sections 1.1–1.3. The aim of this unit is to know what an atom is made of, how we found out, and how to describe a nucleus with the two numbers Z and A .

Concepts in detail

1.1 Imaging atoms: from an idea to an object

Atom. The smallest particle of a chemical element that keeps the identity of that element. The word comes from the Greek *atomos*, “that cannot be cut”.

- **Democritus** (5th century BC): matter cannot be divided forever; there must be indivisible building blocks, the atoms. This was a philosophical idea with no experiment behind it.
- **Dalton** (1808): the first *scientific* atomic theory. Each element is made of one type of atom with its own mass, and a given compound always contains the same ratio of atoms (for example water always has 2 H for 1 O). This explains the law of definite proportions.
- **Scanning tunnelling microscopy** (STM, 1980s): a sharp metal tip is moved a few tenths of a nanometre above a surface; the tiny current that “tunnels” between tip and surface depends very strongly on the distance, so the surface can be mapped atom by atom. The same tip can push atoms around: Figure 1-1 of the handout is an ellipse of cobalt atoms placed one by one on a copper surface. The atom went from a hypothesis to a real object that can be seen and moved.

1.2 Internal structure: the atom is not indivisible

1.2.1 The electron (Thomson, 1897)

A high voltage between two electrodes in a tube containing a gas at low pressure produces a beam coming out of the negative electrode (the cathode): a *cathode ray*.

- The beam is deflected by an electric field *towards the positive plate*: it carries a **negative** charge.
- The deflection is the same whatever the gas and the metal of the electrodes: the particles are a component of *all* atoms.
- From the deflection in electric and magnetic fields, Thomson measured the charge-to-mass ratio $e/m_e \approx 1.76 \times 10^{11}$ C/kg, about 1800 times larger than for the lightest ion (H^+). The particle is therefore much lighter than an atom: the **electron**.

Consequence: atoms contain electrons, and since matter is neutral they must also contain positive charge. Thomson imagined the “plum pudding” model: electrons embedded in a uniform positive sphere.

1.2.2 The nucleus (Rutherford, 1911)

α -particles (He^{2+} ions, positive and heavy) are fired at a very thin gold foil, and a detector all around the foil records where they go.

Observation	Conclusion
Almost all α -particles pass straight through.	The atom is mostly empty space .
A few are deflected by large angles.	They pass close to a small region of concentrated positive charge that repels them.
Very rarely (about 1 in 10 000) one bounces back.	That region is tiny but contains almost all the mass of the atom: the nucleus .

The plum-pudding model cannot explain the bounce-backs: a diffuse positive charge would never stop a fast, heavy α -particle. Rutherford's model: a tiny, dense, positive nucleus surrounded by electrons at comparatively huge distances.

1.2.3 Sizes, charges and masses

- Size of an atom: about $100 \text{ pm} = 10^{-10} \text{ m}$ ($1 \text{ pm} = 10^{-12} \text{ m}$; the older unit $1 \text{ Å} = 100 \text{ pm}$).
- Size of a nucleus: 10^{-3} to 10^{-2} pm , that is 1 to 10 femtometres ($1 \text{ fm} = 10^{-15} \text{ m}$). The atom is 10^4 to 10^5 times larger than its nucleus: if the nucleus were a marble of 1 cm, the atom would be a sphere of about 1 km.
- **Millikan** (1909, oil-drop experiment): every charge he measured was a whole multiple of the same value, the **elementary charge** $e = 1.6 \times 10^{-19} \text{ C}$. Electron: $-e$; proton: $+e$; neutron: 0.
- Masses (Table 1-1): $m_e = 9.1094 \times 10^{-31} \text{ kg}$, $m_p = 1.6727 \times 10^{-27} \text{ kg}$, $m_n = 1.6749 \times 10^{-27} \text{ kg}$. The proton and neutron have almost the same mass, and are about **1836 times** heavier than the electron. Almost all the mass of an atom is in its nucleus.

Atomic mass unit (amu, also written u or Da). Exactly $\frac{1}{12}$ of the mass of one unbound carbon-12 atom, at rest and in its ground state: $1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$. It is chosen so that protons and neutrons have masses close to 1 amu, and it links to the mole: $1 \text{ amu} \times N_A = 1 \text{ g/mol}$.

Worked example: converting Table 1-1 into amu $m_p/(1 \text{ amu}) = 1.6727/1.6605 = 1.0073$; $m_e/(1 \text{ amu}) = 9.1094 \times 10^{-31}/1.6605 \times 10^{-27} = 5.49 \times 10^{-4} = 1/1823$; gold atom: $3.2707 \times 10^{-25}/1.6605 \times 10^{-27} = 196.97$. (For the hydrogen atom the same division gives 1.0078, which rounds to 1.008.)

1.3 Chemical element and isotopes

Atomic number Z and mass number A . Z = number of protons in the nucleus (also the number of electrons in the neutral atom). A = number of nucleons (protons + neutrons). The number of neutrons is $N = A - Z$. A nucleus (or atom) is written ${}_Z^AX$, where X is the symbol of the element, for example ${}_6^{12}\text{C}$, ${}_{17}^{35}\text{Cl}$, ${}_{26}^{56}\text{Fe}$.

Chemical element. The set of all atoms and ions that have the same atomic number Z . Z alone fixes the element: every atom with 6 protons is carbon, whatever its number of neutrons or electrons.

Isotopes. Atoms of the same element (same Z) with different numbers of neutrons, hence different A . Example: ${}_6^{12}\text{C}$ (6 neutrons), ${}_6^{13}\text{C}$ (7) and ${}_6^{14}\text{C}$ (8). Isotopes have the same electrons, hence the same chemistry; only their mass (and nuclear stability) differ.

Ions. An atom that has lost or gained electrons. The nucleus is unchanged. A cation X^{q+} has $Z - q$ electrons; an anion X^{q-} has $Z + q$ electrons. Species with the same number of electrons are called **isoelectronic** (for example F^- , Ne , Na^+ , Mg^{2+} all have 10 electrons).

Common mistake A is a *count* of nucleons (an integer), not the atomic mass. The mass of ${}_{17}^{35}\text{Cl}$ is 34.97 amu, and the atomic mass of chlorine found in tables (35.45) is an average over its isotopes (Unit 5). Also, changing the number of electrons makes an ion, not an isotope.

Recap · Unit 1: classical description of the atom

Timeline

5th c. BC	Democritus	indivisible atoms (philosophy)
1808	Dalton	each element has its own atoms and mass; fixed ratios
1897	Thomson	electron: cathode rays, negative, e/m_e very large
1909	Millikan	charge is quantised: $e = 1.6 \times 10^{-19} \text{ C}$
1911	Rutherford	gold foil: tiny, dense, positive nucleus; the atom is mostly empty
1980s	STM	individual atoms imaged and moved

Particles

	charge	mass (kg)	mass (amu)
electron	$-e$	9.1094×10^{-31}	$1/1823 \approx 0.00055$
proton	$+e$	1.6727×10^{-27}	1.0073
neutron	0	1.6749×10^{-27}	1.0087

$$1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg} \\ = \frac{1}{12} m(^{12}\text{C})$$

$$m_p/m_e \approx 1836$$

$$1 \text{ amu} \times N_A = 1 \text{ g mol}^{-1}$$

Orders of magnitude

Atom $\sim 100 \text{ pm} = 10^{-10} \text{ m}$; nucleus $\sim 10^{-3}\text{--}10^{-2} \text{ pm}$ (1–10 fm); ratio $10^4\text{--}10^5$. The nucleus holds $> 99.9\%$ of the mass.

Notation and counting

${}_Z^AX$: protons $= Z$, neutrons $N = A - Z$, electrons $= Z - q$ for charge q (so $Z + |q|$ for an anion).

Element $\Leftrightarrow$ same Z . Isotopes $\Leftrightarrow$ same Z , different A . Isoelectronic $\Leftrightarrow$ same number of electrons.

You should be able to

- describe the Thomson and Rutherford experiments and say what each observation proves;
- convert between pm, nm, Å, fm and m, and between kg and amu;
- give Z , A , N and the electron count of any atom or ion written ${}_Z^AX^{q\pm}$;
- recognise isotopes and isoelectronic species.

Watch out

- Cathode rays bend *towards* the positive plate (they are negative).
- A cation has *fewer* electrons than protons; an anion has *more*.
- A is an integer count; masses in amu are not integers (except ^{12}C , by definition).

Worksheet 1 · Classical description of the atom

Data $e = 1.6 \times 10^{-19} \text{ C}$ $m_e = 9.1094 \times 10^{-31} \text{ kg}$ $m_p = 1.6727 \times 10^{-27} \text{ kg}$ $m_n = 1.6749 \times 10^{-27} \text{ kg}$
 $1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$ $N_A = 6.022 \times 10^{23} / \text{mol}$ $c = 3.00 \times 10^8 \text{ m/s}$ $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Calculator allowed.

Q1. Level 1 • Who discovered what?

Match each scientist to the date and to the discovery or idea: Democritus, Dalton, Thomson, Millikan, Rutherford. Discoveries: (a) the nucleus; (b) the elementary charge; (c) indivisible atoms; (d) the electron; (e) compounds contain fixed ratios of atoms of given masses.

Q2. Level 1 • Counting particles

Give the number of protons, neutrons and electrons of $^{12}_6\text{C}$, $^{14}_6\text{C}$, $^{35}_{17}\text{Cl}$ and $^{56}_{26}\text{Fe}$. Which two are isotopes?

Q3. Level 1 • Units of length

Express a typical atomic size, 100 pm, in m, nm and Å. Express 1 fm in pm. What is the typical ratio between the size of an atom and the size of its nucleus?

Q4. Level 1 • Cathode rays

In Thomson's tube the beam bends towards the positively charged plate. What does this show? Why was it important that the same deflection was obtained with any gas and any metal?

Q5. Level 2 •• Ions and isoelectronic species

Give the number of protons, neutrons and electrons of $^{24}_{12}\text{Mg}^{2+}$, $^{27}_{13}\text{Al}^{3+}$, $^{19}_9\text{F}^-$ and $^{40}_{20}\text{Ca}^{2+}$. Which species are isoelectronic, and with which noble gas?

Q6. Level 2 •• Reading Rutherford's experiment

Give the three observations of the gold-foil experiment and, for each one, what it tells us about the atom. Why could Thomson's "plum pudding" model not explain them?

Q7. Level 2 •• How heavy are electrons?

(a) Compute m_p/m_e . (b) What fraction of the mass of a ^{12}C atom (exactly 12 amu) is due to its electrons? Conclude.

Q8. Level 2 •• A stadium-sized atom

Take an atom of diameter 100 pm and a nucleus of diameter 10^{-3} pm. If the nucleus were enlarged to a marble of diameter 1.0 cm, what would be the diameter of the atom? What fraction of the atom's volume does the nucleus occupy?

Q9. Level 3 ••• Charge-to-mass ratios

(a) Compute e/m for the electron and for the proton. (b) An electron and a proton enter the same electric field with the same speed. Which one is deflected more, and by what factor (the deflection is proportional to q/m)? (c) Why did this make the electron easy to discover with cathode rays?

Q10. Level 3 ••• From amu to grams per mole

One atom of gold has a mass of 196.967 amu. (a) Express it in kg. (b) How many atoms are there in 1.00 g of gold? (c) How many moles is that? Deduce the molar mass of gold and state the general rule linking amu and g mol^{-1} .

Q11. Level 3 ●●● Identify the species

Identify each species (symbol, A , charge). Use a periodic table for the symbols. (a) 26 protons, 30 neutrons, 23 electrons. (b) 17 protons, 20 neutrons, 18 electrons. (c) A neutral atom with $A = 39$ and one more neutron than protons. (d) An ion with 10 electrons, 8 neutrons and charge $2-$.

Q12. Level 3 ●●● Density of nuclear matter

A ^{12}C nucleus has a mass of about 12 amu and a radius of about 2.7 fm. (a) Compute its density. (b) What would be the mass of a teaspoon (5 cm^3) of nuclear matter? Compare with water (1000 kg/m^3).

Q13. Challenge ★ Millikan's oil drops

Millikan measured the charges of oil drops: 3.2×10^{-19} , 4.8×10^{-19} , 8.0×10^{-19} , 6.4×10^{-19} and $1.12 \times 10^{-18}\text{ C}$. (a) Show that these values are consistent with a single elementary charge and find it. (b) How many extra electrons does each drop carry? (c) Could a drop carry $2.4 \times 10^{-19}\text{ C}$?

Q14. Challenge ★ Where does the missing mass go?

(a) Using the masses in amu of the proton (1.0073), the neutron (1.0087) and the electron ($1/1823$), compute the total mass of the particles that make up a ^{12}C atom. (b) Compare with the mass of the atom, 12 amu exactly: compute the mass defect Δm in amu and in kg. (c) The energy that holds the nucleus together is $E = \Delta m c^2$. Compute it per atom and per mole, and compare with a typical chemical bond energy (about 400 kJ mol^{-1}). Why can chemical reactions not change one element into another?

Q15. Challenge ★ How small is the gold nucleus?

In Rutherford's experiment, an α -particle (charge $+2e$) with kinetic energy 5.0 MeV ($1\text{ MeV} = 1.6 \times 10^{-13}\text{ J}$) heads straight for a gold nucleus ($Z = 79$). It stops and turns back at the distance d where all its kinetic energy has become electric potential energy $E_p = k q_1 q_2 / d$. (a) Compute d . (b) What does this tell you about the size of the nucleus? Compare with the size of the gold atom (about 140 pm).

UNIT 2

Quantum model of the atom

Handout chapter 2, sections 2.1–2.5. Light is both a wave and a stream of photons; atoms absorb and emit light only at particular wavelengths; Bohr explains this with quantised energy levels.

Concepts in detail

2.1 Electromagnetic radiation

Energy. Energy is the capacity of a system to produce work or heat. It exists in several forms that can be converted into one another; electromagnetic (EM) radiation is one of them.

Completing the handout: Table 2-1, the 8 forms of energy

The usual classification has two families of four. The top wide row is “kinetic energy” (energy of motion), the bottom wide row “potential energy” (stored energy).

Kinetic energy (motion of waves, particles, objects)			
Radiant (EM radiation: light, X-rays, radio...)	Thermal (random motion of particles)	Motion (mechanical, of a moving object)	Sound (vibration of matter)
Potential energy (stored, due to position or structure)			
Chemical (in bonds between atoms)	Nuclear (in the nucleus)	Elastic (stored mechanical: spring, stretched material)	Gravitational (position in a gravity field)

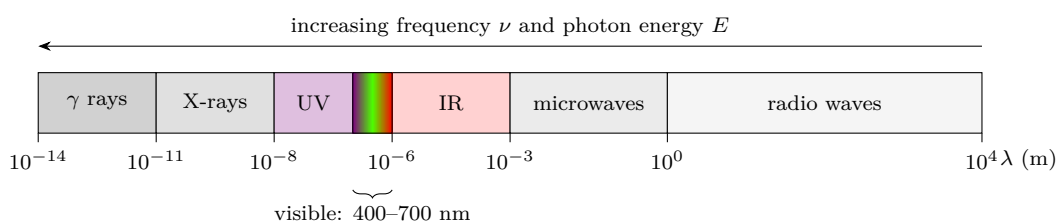
Some textbooks list *electrical* energy instead of sound or elastic energy; use the list given in class if it differs. The point for this course: EM radiation is *radiant* energy.

EM radiation as a wave. Coupled oscillations of an electric field and a magnetic field, perpendicular to each other and to the direction of travel. In a vacuum it travels at $c = 3.00 \times 10^8$ m/s, whatever its wavelength.

Quantity	Definition	SI unit
Temporal period T	duration of one oscillation	s
Wavelength λ	distance travelled during one period (spatial period)	m
Frequency ν	number of oscillations per second	Hz = s ⁻¹

$$\boxed{\lambda = cT} \quad \boxed{\nu = \frac{1}{T}} \quad \Rightarrow \quad \boxed{\lambda = \frac{c}{\nu}} \quad \text{or} \quad \nu = \frac{c}{\lambda}.$$

Short wavelength means high frequency. The **EM spectrum** orders the radiations by wavelength:



Visible light goes from violet (about 400 nm) to red (about 700 nm): violet < blue < green < yellow < orange < red in wavelength. UV is just below 400 nm, IR just above 700 nm.

2.2 Quantisation: the photon

Photon. A particle of EM radiation with no mass, travelling at c . Its energy is fixed by the frequency (Planck):

$$E = h\nu = \frac{hc}{\lambda} \quad h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s (Planck's constant)}.$$

Wave–particle duality. EM radiation behaves like a wave in some experiments (diffraction, interference) and like a stream of particles in others (absorption and emission of energy in packets $h\nu$). Both descriptions are needed.

Electronvolt. The energy gained by one electron accelerated through 1 V: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. Atomic energies are a few eV. Handy shortcut: $E (\text{eV}) \approx \frac{1240}{\lambda (\text{nm})}$ (with the handout's constants, $hc/e = 1241 \text{ eV nm}$).

Worked example: a red photon $\lambda = 656 \text{ nm}$: $\nu = c/\lambda = 3.00 \times 10^8 / 656 \times 10^{-9} = 4.57 \times 10^{14} \text{ Hz}$; $E = h\nu = 6.62 \times 10^{-34} \times 4.57 \times 10^{14} = 3.03 \times 10^{-19} \text{ J} = 1.89 \text{ eV}$.

Common mistake Convert nm to m before using $\lambda = c/\nu$ (1 nm = 10^{-9} m), and joules to eV by *dividing* by 1.6×10^{-19} . A blue photon has *more* energy than a red one.

2.3 Interaction between EM radiation and matter

- **Reflection:** the radiation bounces back at a surface. **Refraction:** it changes direction when it enters another medium, because it travels at a different speed there.
- **Scattering:** the radiation is re-emitted in all directions by particles (why the sky is blue: short wavelengths are scattered most).
- **Absorption:** the energy of the photon is taken up by the matter. Only photons whose energy matches a possible change of energy of the matter are absorbed.

Absorption spectrum and colour. The spectrum of the light that has passed through a sample; missing wavelengths appear black. A solution of KMnO_4 absorbs a *broad band* (about 515–565 nm, green): molecules in a solution have very many closely spaced energy levels, so the absorbed wavelengths form a continuous band. What we see is the transmitted light, mainly red and violet–blue: the solution looks purple (magenta), the **complementary colour** of the absorbed green.

Note: on Figures 2-2 to 2-4 of the handout, the right-hand axis label “650” is a typo for 700.

2.4 Atomic spectra

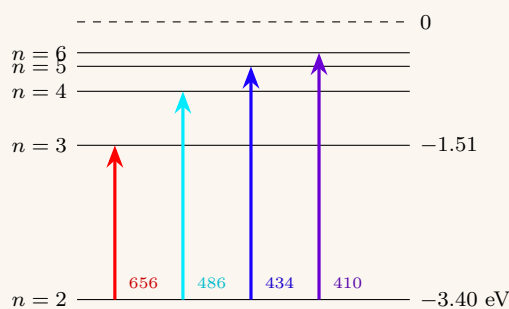
2.4.1 Absorption spectrum of hydrogen

White light through hydrogen gas: the spectrum shows only four thin black lines in the visible, at 656, 486, 434 and 410 nm. The hydrogen atom absorbs only certain wavelengths, so it can only take in certain amounts of energy: its energy is **quantised**.

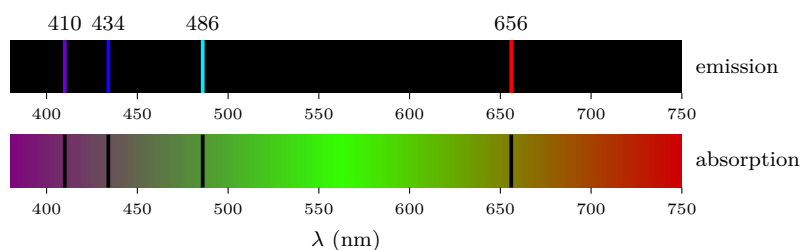
Completing the handout: box of 2.4.1, why only a few lines?

A photon is absorbed only if its energy is exactly the difference between two energy levels of the atom: $h\nu = E_p - E_n$. The atom has a discrete set of levels, so only a discrete set of wavelengths is absorbed. The four visible lines all start from level $n = 2$ and go to $p = 3, 4, 5, 6$ (arrows on the right). Transitions from $n = 1$ need more energy (UV) and those from $n = 3$ less (IR); they are outside the visible window.

Note: at room temperature almost every H atom is in $n = 1$, so the visible absorption lines are seen when the gas is hot (in a discharge, or in the atmosphere of a star); the important idea is the match between photon energy and level spacing.

**2.4.2 Emission spectrum of hydrogen**

An electric discharge excites hydrogen gas, which then emits light. Its spectrum is a black background with four coloured lines at exactly the same wavelengths as the absorption lines:



The same lines appear in emission and absorption because the same pairs of levels are involved: in emission the electron goes *down* and a photon is produced; in absorption a photon is used to push the electron *up*. The spectrum is a fingerprint of the element.

2.4.3 Spectral series: Balmer and Rydberg

Balmer (1885) found that the frequencies of the four visible lines follow $\nu \propto \frac{1}{4} - \frac{1}{n^2}$ with $n = 3, 4, 5, 6$ (the handout writes $n = 1, 2, 3 \dots$, but n must be larger than 2 for the frequency to be positive). Rydberg generalised this to every series:

$$\nu = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad n_1 < n_2 \text{ positive integers, } R_H = 3.29 \times 10^{15} \text{ Hz.}$$

Each value of n_1 gives a **series**; the lines of a series crowd together towards the **series limit** ($n_2 \rightarrow \infty$, $\nu = R_H/n_1^2$).

Series	n_1	Region	Wavelengths
Lyman	1	UV	122 nm ($2 \rightarrow 1$) down to the limit 91.2 nm
Balmer	2	visible (and near UV)	656, 486, 434, 410 nm ... limit 365 nm
Paschen	3	IR	1876 nm ($4 \rightarrow 3$) ... limit 821 nm
Brackett, Pfund	4, 5	far IR	

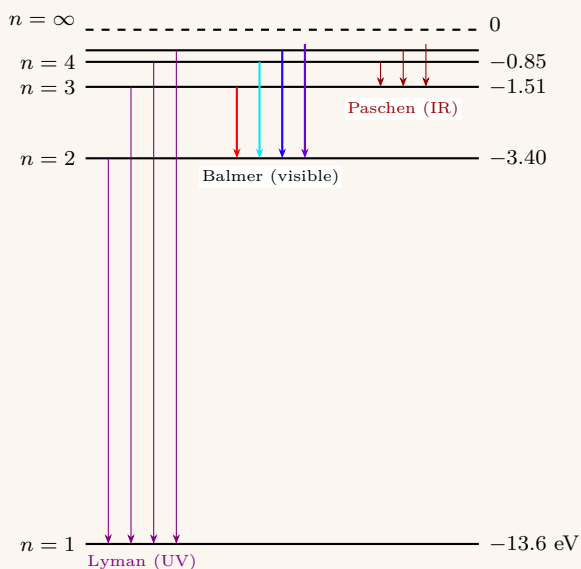
2.5 Electronic energy levels: Bohr's model (1913)

Bohr's two postulates. (1) The electron of a hydrogen atom can only have certain energies:

$E_n = -\frac{13.6}{n^2} \text{ eV}$, $n = 1, 2, 3, \dots$: its energy is **quantised**. (2) The atom emits or absorbs light only when the electron makes a **transition** between two levels; the photon carries the difference of energy.

- **Ground state:** $n = 1$, $E_1 = -13.6$ eV, lowest energy, strongest interaction with the proton.
- **Excited states:** $n > 1$ ($E_2 = -3.40$, $E_3 = -1.51$, $E_4 = -0.85$ eV ...), higher energy, weaker interaction.
- **Ionised state:** $n \rightarrow \infty$, $E_\infty = 0$: the electron is free. The energies are negative because the bound electron has *less* energy than a free one at rest.
- **Ionisation energy** of H from the ground state: $E_\infty - E_1 = 13.6$ eV.

Completing the handout: side box of 2.5, energy-level diagram of hydrogen

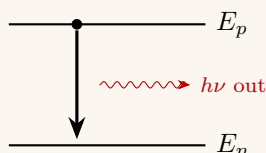


Each horizontal line is an allowed energy $E_n = -13.6/n^2$ eV. The levels get closer and closer as n increases and pile up under $E = 0$ (ionisation). Each downward arrow is an emission line. All arrows ending on the same level form one series: on $n = 1$ the Lyman series (large gaps, UV), on $n = 2$ the Balmer series (visible), on $n = 3$ the Paschen series (IR).

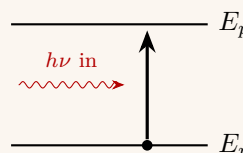
Rydberg's formula is Bohr's model in disguise: $h\nu = 13.6 \text{ eV} (1/n_1^2 - 1/n_2^2)$, so $R_H = 13.6 \text{ eV}/h = 13.6 \times 1.6 \times 10^{-19} / 6.62 \times 10^{-34} = 3.29 \times 10^{15} \text{ Hz}$.

Completing the handout: table after 2.5, emission and absorption

Emission: an electron on level E_p falls to a lower level E_n and a photon is emitted.



Absorption: an electron on level E_n takes in a photon and is promoted to E_p .



In both cases the photon carries exactly the energy difference: $\Delta E = E_p - E_n = h\nu = \frac{hc}{\lambda}$. A photon whose energy does not match any gap is not absorbed (unless it has enough energy to ionise the atom; the excess then becomes kinetic energy of the free electron).

Worked example: the red Balmer line from Bohr's levels $3 \rightarrow 2$: $\Delta E = E_3 - E_2 = -1.51 - (-3.40) = 13.6 (\frac{1}{4} - \frac{1}{9}) = 1.89 \text{ eV} = 1.89 \times 1.6 \times 10^{-19} = 3.02 \times 10^{-19} \text{ J}$, so $\lambda = hc/\Delta E = 6.62 \times 10^{-34} \times 3.00 \times 10^8 / 3.02 \times 10^{-19} = 657 \text{ nm}$, the observed 656 nm line.

Hydrogen-like ions. A nucleus of charge Z with a single electron (He^+ , Li^{2+} ...): $E_n = -13.6 Z^2/n^2$ eV. The electron is held Z^2 times more strongly.

Recap · Unit 2: quantum model of the atom**Key equations**

Wave $\lambda = cT, \quad \nu = 1/T, \quad \lambda = c/\nu$

Photon (Planck) $E = h\nu = hc/\lambda$

Bohr levels (H) $E_n = -13.6/n^2 \text{ eV}; \quad \text{hydrogen-like ion: } E_n = -13.6 Z^2/n^2 \text{ eV}$

Transition $\Delta E = E_p - E_n = h\nu = hc/\lambda \quad (p > n)$

Rydberg $\nu = R_H (1/n_1^2 - 1/n_2^2), \quad R_H = 3.29 \times 10^{15} \text{ Hz} = 13.6 \text{ eV}/h$

Constants and conversions

$c = 3.00 \times 10^8 \text{ m s}^{-1}, \quad h = 6.62 \times 10^{-34} \text{ J s}, \quad 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, \quad E(\text{eV}) \approx 1240/\lambda(\text{nm}),$
 $1 \text{ nm} = 10^{-9} \text{ m}.$

Key numbers for hydrogen

$E_1 = -13.6, E_2 = -3.40, E_3 = -1.51, E_4 = -0.85, E_\infty = 0 \text{ eV}$. Visible lines (Balmer, to $n = 2$): 656 (red, $3 \rightarrow 2$), 486 (blue-green, $4 \rightarrow 2$), 434 (violet, $5 \rightarrow 2$), 410 nm (violet, $6 \rightarrow 2$). Lyman to $n = 1$: UV; Paschen to $n = 3$: IR. Ionisation energy of H: 13.6 eV, i.e. 1310 kJ mol^{-1} (tabulated: 1312).

Vocabulary

wavelength, frequency, period; photon; wave-particle duality; absorption / emission spectrum; line vs band spectrum; quantised energy; transition; ground, excited, ionised state; series and series limit; complementary colour.

You should be able to

- convert between λ, ν, T and E (in J and eV), and place a radiation on the spectrum;
- explain why atoms give line spectra and why emission and absorption lines coincide;
- compute E_n , the energy and wavelength of any transition of H or of a hydrogen-like ion;
- decide whether a photon of given energy is absorbed by a hydrogen atom.

Watch out

- E_n is negative; ΔE for a photon is always taken positive.
- Short λ = high ν = high energy (UV > visible > IR).
- In Rydberg's formula n_1 is the *lower* level: $n_1 < n_2$.

Worksheet 2 · Quantum model of the atom

Data $c = 3.00 \times 10^8 \text{ m/s}$ $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $N_A = 6.022 \times 10^{23} / \text{mol}$
 $R_H = 3.29 \times 10^{15} \text{ Hz}$ hydrogen: $E_n = -13.6/n^2 \text{ eV}$; hydrogen-like ion: $E_n = -13.6 Z^2/n^2 \text{ eV}$. Calculator allowed.

Q1. Level 1 • Ordering the spectrum

Put these radiations in order of increasing frequency: visible light, X-rays, radio waves, infrared, γ rays, ultraviolet, microwaves. Which one has the longest wavelength? Which photons carry the most energy?

Q2. Level 1 • A red laser

A laser emits red light at $\lambda = 650 \text{ nm}$. Compute its frequency ν and its period T .

Q3. Level 1 • Energy of one photon

Compute the energy of one photon of the laser of Q2, in joules and in electronvolts.

Q4. Level 1 • Vocabulary of the Bohr model

Define ground state, excited state and ionised state for the hydrogen atom, and give the energy of the electron in each case (for the excited state, give E_2 and E_3).

Q5. Level 2 •• Emission versus absorption

Describe the emission spectrum and the absorption spectrum of hydrogen gas in the visible. Explain why the lines appear at the same wavelengths in both.

Q6. Level 2 •• Why is permanganate purple?

An aqueous solution of KMnO_4 absorbs light between about 515 and 565 nm. (a) Which colour is absorbed? What colour does the solution appear? (b) Why is the absorption a broad band and not thin lines as for hydrogen gas?

Q7. Level 2 •• The Balmer lines from Rydberg's formula

Using $\nu = R_H(1/n_1^2 - 1/n_2^2)$ with $n_1 = 2$, compute the wavelengths for $n_2 = 3, 4, 5, 6$ and compare them with the handout values.

Q8. Level 2 •• Rydberg meets Bohr

(a) Using Bohr's levels, compute the energy (in eV and J) of the photon emitted in the transition $3 \rightarrow 2$, and its wavelength. (b) Show that Bohr's formula gives Rydberg's formula and that $R_H = 13.6 \text{ eV}/h$. Check the numerical value.

Q9. Level 3 ••• Series and their limits

(a) Compute the wavelength of the first Lyman line ($2 \rightarrow 1$) and the Lyman series limit. In which region are they? (b) Compute the Paschen series limit ($n_1 = 3$). (c) Explain why only the Balmer series has lines in the visible.

Q10. Level 3 ••• Ionisation energy of hydrogen

(a) What energy is needed to ionise a hydrogen atom in its ground state? Give it in eV, in J per atom and in kJ mol^{-1} . (b) What is the longest wavelength that can do it? (c) Unit 5 lists ionisation energies in kJ mol^{-1} but leaves hydrogen blank: fill it in.

Q11. Level 3 ●●● Counting emission lines

Hydrogen atoms are excited to $n = 4$ and then return to the ground state by all possible paths. (a) How many different lines can be emitted? List the transitions. (b) Sort them into series and spectral regions. (c) Which ones can you see?

Q12. Level 3 ●●● Which photons are absorbed?

Hydrogen atoms in their ground state are hit by photons of energy (a) 10.2 eV, (b) 11.0 eV, (c) 12.09 eV, (d) 15.0 eV. For each, say whether it is absorbed and what happens to the atom.

Q13. Challenge ★ Identify the transition

A line of the hydrogen emission spectrum is measured at $\lambda = 1876$ nm. Find n_1 and n_2 . Explain your method.

Q14. Challenge ★ The helium ion

He^+ ($Z = 2$) has a single electron. (a) Give its energy levels E_1 , E_2 , E_3 and its ionisation energy. (b) Compute the wavelength of its $3 \rightarrow 2$ transition; in which region is it? (c) Show that the He^+ transition $6 \rightarrow 4$ has exactly the same wavelength as the red line of hydrogen. Why is this so?

Q15. Challenge ★ Photons in a laser beam

A green laser pointer emits 1.0 mW at 532 nm. (a) How many photons does it emit per second? (b) How many moles of photons per hour? (c) Why do we not notice that light comes in packets?

UNIT 3

Quantum numbers and atomic orbitals

Handout chapter 3, sections 3.1–3.4. The electron is described by a wavefunction labelled by quantum numbers; each set of quantum numbers is an atomic orbital with a given energy, size, shape and orientation.

Concepts in detail

3.1 The nature of the electron

Wave–particle duality of the electron. Like light, the electron is both a particle (it has a mass and a charge) and a wave (a beam of electrons is diffracted by a crystal, like X-rays).

Consequence: the electron does not follow a trajectory (no orbit as in a solar system). We can only give the **probability** of finding it at each point around the nucleus. The classical picture of Figure 3-1 (an ellipse) is replaced by an electron *cloud*, dense where the probability is high.

3.2 The hydrogen electron and its quantum numbers n, ℓ, m_ℓ

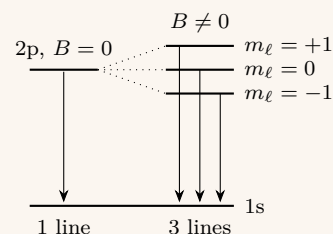
The wavefunction Ψ of the single electron of hydrogen is fixed by three whole numbers:

Quantum number	Allowed values	Fixes	Name of the group
principal n	$1, 2, 3, \dots$	energy (in H) and size	shell
secondary (azimuthal) ℓ	$0 \leq \ell \leq n - 1$	shape	subshell
magnetic m_ℓ	$-\ell \leq m_\ell \leq +\ell$ (integer)	orientation in space	one orbital

- For a given n there are n values of ℓ ; for a given ℓ there are $2\ell + 1$ values of m_ℓ ; a shell contains n^2 orbitals in total.
- Subshells are named by letters: $\ell = 0$: **s**; $\ell = 1$: **p**; $\ell = 2$: **d**; $\ell = 3$: **f**. The name is n followed by the letter: 1s, 2p, 3d, 4f.
- The $2\ell + 1$ orbitals of a subshell point in different directions but have the same energy (they are *degenerate*), as long as no magnetic field is applied.

Completing the handout: box after 3.2.3, the Zeeman effect

In a magnetic field, the orbitals of one subshell no longer have the same energy: each value of m_ℓ gets a slightly different energy. A p level ($\ell = 1$) splits into $2\ell + 1 = 3$ sublevels ($m_\ell = -1, 0, +1$), a d level into 5. A transition such as $2p \rightarrow 1s$ then gives three close lines instead of one: the line is *split*. Zeeman's observation is the experimental proof that m_ℓ exists and takes $2\ell + 1$ values.



3.3 Atomic orbitals of hydrogen

Atomic orbital (AO). The wavefunction Ψ_{n,ℓ,m_ℓ} defined by one set of three quantum numbers, and by extension the region of space where the electron described by it is likely to be found. $|\Psi|^2$ is the probability density of presence of the electron.

Completing the handout: Table 3-1, quantum numbers, wavefunctions and atomic orbitals

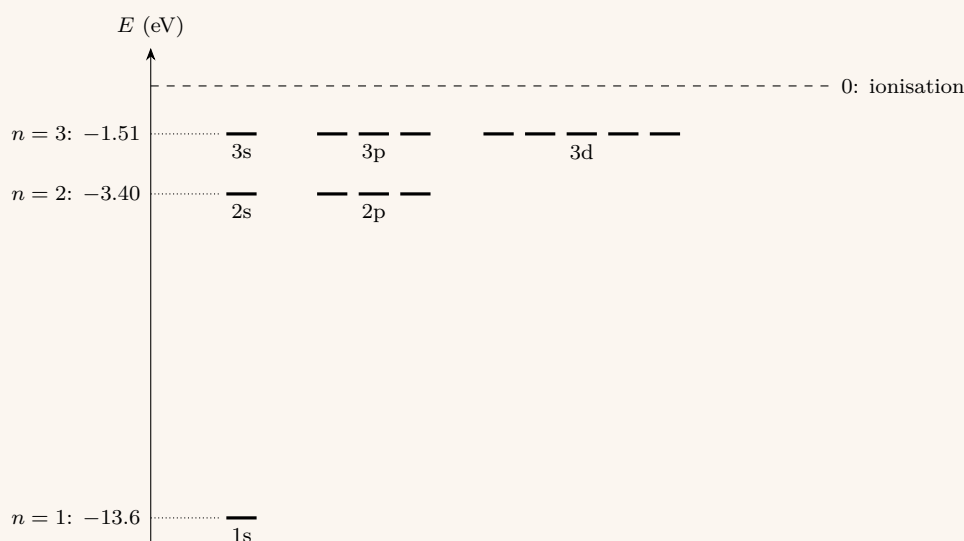
n	E (eV)	ℓ	m_ℓ	wavefunction	atomic orbital
1	-13.6	0	0	$\Psi_{1,0,0}$	1s
2	-3.40	0	0	$\Psi_{2,0,0}$	2s
		1	-1	$\Psi_{2,1,-1}$	2p (3 AOs)
			0	$\Psi_{2,1,0}$	
			+1	$\Psi_{2,1,1}$	
3	-1.51	0	0	$\Psi_{3,0,0}$	3s
		1	-1	$\Psi_{3,1,-1}$	3p (3 AOs)
			0	$\Psi_{3,1,0}$	
			+1	$\Psi_{3,1,1}$	
		2	-2	$\Psi_{3,2,-2}$	3d (5 AOs)
			-1	$\Psi_{3,2,-1}$	
			0	$\Psi_{3,2,0}$	
			+1	$\Psi_{3,2,1}$	
			+2	$\Psi_{3,2,2}$	

Count: shell 1 has 1 AO, shell 2 has 4, shell 3 has 9: n^2 AOs per shell.

3.3.1 Energy levels of the atomic orbitals of hydrogen

In hydrogen the energy depends *only* on n : $E_n = -13.6/n^2$ eV. All AOs of the same shell are **degenerate** (2s and 2p have the same energy; 3s, 3p and 3d too).

Completing the handout: Figure 3-2, energy diagram of the AOs of hydrogen (one stroke = one AO)



The scale is linear in energy: the gap $1 \rightarrow 2$ (10.2 eV) is much larger than $2 \rightarrow 3$ (1.89 eV).

Hydrogen-like ions. $E_n = -13.6 Z^2/n^2$ eV. $E = 0$ means no interaction between the nucleus and the electron.

3.3.2 Occupied and unoccupied orbitals

An AO is **occupied** when the electron is described by that wavefunction; all the others are **unoccupied** (empty, but they still exist and can receive the electron after absorption of a photon). Hydrogen is in its ground state when the electron occupies 1s, in an excited state otherwise.

Completing the handout: Table 3-2, occupied and unoccupied AOs

Ψ_{n,ℓ,m_ℓ}	1s	2p	3d	state
$\Psi_{1,0,0}$	occupied	unoccupied	unoccupied	ground state
$\Psi_{3,2,-1}$	unoccupied	unoccupied	occupied	excited state

$\Psi_{3,2,-1}$: $n = 3$, $\ell = 2$ (d), so the electron is in a 3d orbital, with energy -1.51 eV instead of -13.6 eV.

3.3.3 Shapes of the atomic orbitals

Boundary surface. A surface that encloses a given probability of finding the electron, usually 95 %. Its shape is the “shape of the orbital” drawn in books.

Radial probability density. The probability of finding the electron at a distance between r and $r + dr$ from the nucleus, per unit dr (Figure 3-3). Its maximum gives the most probable distance.

Bohr radius a_0 . The most probable distance between the nucleus and the electron of hydrogen in its ground state (1s): $a_0 = 52.9$ pm. The size of an AO of a hydrogen-like species grows with n

and shrinks with Z : $r = \frac{n^2}{Z} a_0$.

Nodes and phase. A **node** is a place where $\Psi = 0$ (zero probability). Crossing a node, the wavefunction changes sign (phase shift of π); the two signs are shown with two colours or $+/ -$ labels. An AO has $n - \ell - 1$ **radial nodes** (spheres) and ℓ **angular nodes** (planes or cones): $n - 1$ nodes in total.

AO	1s	2s	2p	3s	3p	3d	4s
radial nodes $n - \ell - 1$	0	1	0	2	1	0	3
angular nodes ℓ	0	0	1	0	1	2	0

You can check the radial nodes on Figure 3-3: the curve of 3s touches zero twice before its main peak, 3p once, 3d never.

- **s orbitals** ($\ell = 0$): spherical. They get larger as n increases, and ns has $n - 1$ spherical nodes inside (the “onion layers” of Figure 3-4).
- **p orbitals** ($\ell = 1$): two lobes of opposite sign on each side of the nucleus (“diabolo” or dumbbell), separated by a **nodal plane** through the nucleus. The three p orbitals point along x , y and z .
- **d orbitals** ($\ell = 2$): mostly four lobes (cloverleaf) with two nodal surfaces; they appear in Units 4–5 through the transition metals.

Completing the handout: Table 3-3, spatial orientation of the p orbitals

The columns (xOy), (xOz), (yOz) ask whether each plane is a **nodal plane** of the orbital (the plane where $\Psi = 0$, between the two lobes).

AO	m_ℓ	(xOy)	(xOz)	(yOz)	spatial orientation
p_x	± 1 (combination)	no	no	nodal plane	along the x axis
p_y	± 1 (combination)	no	nodal plane	no	along the y axis
p_z	0	nodal plane	no	no	along the z axis

Only p_z corresponds to a single value, $m_\ell = 0$. The orbitals with $m_\ell = +1$ and -1 are complex; p_x and p_y are the two real combinations of them. Some courses simply write p_x : $+1$ and p_y : -1 ; follow the convention used in class.

3.4 Many-electron atoms

With two or more electrons, the electrons repel each other and the equations can no longer be solved exactly: approximations are needed.

Orbital approximation. Electron–electron interactions are treated as small compared with the nucleus–electron attraction. The wavefunction of N electrons becomes a product of one-electron AOs, $\psi(1, 2, \dots, N) = \psi(1)\psi(2)\cdots\psi(N)$: each electron occupies its own AO, **independently** of the others, and the AOs keep the labels n, ℓ, m_ℓ of hydrogen.

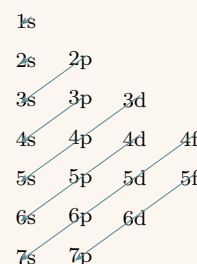
Energy of AOs in a many-electron atom. The energy now depends on n, ℓ and Z : within a shell, $ns < np < nd$. The degeneracy of a shell is lifted, but the $2\ell + 1$ orbitals of a subshell remain degenerate.

Why? The other electrons partly *screen* the nucleus. An s electron spends some time very close to the nucleus (it *penetrates* the inner shells, see the small inner peaks of 2s and 3s on Figure 3-3), so it feels a larger charge and is more strongly bound than a p electron of the same shell, which is itself more strongly bound than a d electron.

Madelung's rule ($n + \ell$ rule). (1) The energy of an AO increases with $n + \ell$. (2) For equal $n + \ell$, the energy increases with n .

Completing the handout: box after Madelung's rule, order of the subshells

$n + \ell$	subshells (increasing energy)
1	1s
2	2s
3	2p, 3s
4	3p, 4s
5	3d, 4p, 5s
6	4d, 5p, 6s
7	4f, 5d, 6p, 7s
8	5f, 6d, 7p



Order: $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p$.
The key surprise: 4s ($n + \ell = 4$) is below 3d ($n + \ell = 5$).

Follow the arrows from the top, one after the other.

3.4.3 Spin: the fourth quantum number

Spin magnetic quantum number m_s . The electron has an intrinsic angular momentum, its **spin**. Its projection has only two values: $m_s = +\frac{1}{2}$ (drawn $\uparrow$, “spin up”) or $m_s = -\frac{1}{2}$ ($\downarrow$, “spin down”). An electron in an atom is therefore described by **four** quantum numbers (n, ℓ, m_ℓ, m_s).

Completing the handout: after the sodium D line, what the doublet shows

The yellow light of sodium comes from the transition $3p \rightarrow 3s$ of its outer electron. Without spin, the 3p level would be a single level and give one line. The spin of the electron interacts with its orbital motion around the nucleus (*spin–orbit coupling*); the two orientations of the spin give two 3p levels with slightly different energies, hence two lines:

$$E = \frac{hc}{\lambda} : \quad \frac{hc}{589.0 \text{ nm}} = 2.1074 \text{ eV}, \quad \frac{hc}{589.6 \text{ nm}} = 2.1052 \text{ eV}, \quad \Delta E \approx 2.1 \times 10^{-3} \text{ eV}.$$

The splitting is a thousand times smaller than the transition energy, but it is well resolved in a spectrometer. It is an experimental proof that a fourth quantum number exists.

Common mistake ℓ goes up to $n - 1$, not n (there is no 2d or 1p orbital). m_ℓ is an integer between $-\ell$ and $+\ell$; m_s is a half-integer and can only be $\pm\frac{1}{2}$. In hydrogen 2s and 2p have the same energy; in every other atom 2s is lower.

Recap · Unit 3: quantum numbers and atomic orbitals**The four quantum numbers**

n	$1, 2, 3, \dots$	shell: energy (H) and size	n^2 AOs per shell
ℓ	$0, 1, \dots, n-1$	subshell: shape (s, p, d, f)	$2\ell + 1$ AOs per subshell
m_ℓ	$-\ell, \dots, +\ell$	orientation of the AO	
m_s	$+\frac{1}{2}$ ($\uparrow$) or $-\frac{1}{2}$ ($\downarrow$)	spin	$2n^2$ electron states per shell

Key formulas

Hydrogen and hydrogen-like ions: $E_n = -13.6 Z^2/n^2$ eV (depends on n only: all AOs of a shell are degenerate). Size: $r = n^2 a_0 / Z$, $a_0 = 52.9$ pm. Nodes: $n - \ell - 1$ radial, ℓ angular, $n - 1$ in total.

Shapes

s: sphere; p: two lobes along x , y or z with a nodal plane (p_x : plane yOz , p_y : xOz , p_z : xOy); d: mostly four lobes. Boundary surface = 95 % probability. Sign change across a node.

Many-electron atoms

Orbital approximation: each electron in its own hydrogen-like AO. Energy depends on n and ℓ (and Z). Madelung: increasing $n + \ell$, then increasing n :

1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p.

Experiments

Electron diffraction $\Rightarrow$ the electron is a wave. Zeeman effect (lines split in a magnetic field) $\Rightarrow m_\ell$, $2\ell + 1$ orientations. Sodium D doublet (589.0 and 589.6 nm) $\Rightarrow$ spin, $m_s = \pm \frac{1}{2}$.

You should be able to

- list the allowed quantum numbers of a shell, name the AOs and count them;
- say whether a set (n, ℓ, m_ℓ, m_s) is allowed and name the corresponding AO;
- give the energy and the size of an AO of H or of a hydrogen-like ion;
- count the nodes of an AO and describe s and p boundary surfaces;
- order subshells with Madelung's rule.

Worksheet 3 · Quantum numbers and atomic orbitals

Data $c = 3.00 \times 10^8 \text{ m/s}$ $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $a_0 = 52.9 \text{ pm}$ hydrogen-like species: $E_n = -13.6 Z^2/n^2 \text{ eV}$, $r = n^2 a_0/Z$. Calculator allowed.

Q1. Level 1 • The third shell

For $n = 3$, list the allowed values of ℓ with the name of each subshell, and the values of m_ℓ for each. How many orbitals does the shell contain?

Q2. Level 1 • Allowed or not?

Which of these sets (n, ℓ, m_ℓ) are allowed? Name the orbital when it is, and give the reason when it is not: (a) $(2, 2, 0)$; (b) $(3, 1, -1)$; (c) $(1, 0, 1)$; (d) $(4, 3, -2)$; (e) $(2, 1, 2)$; (f) $(5, 0, 0)$.

Q3. Level 1 • Naming wavefunctions

Name the orbital described by $\Psi_{2,0,0}$, $\Psi_{3,1,1}$, $\Psi_{4,2,-2}$ and $\Psi_{4,3,0}$. Conversely, give all the wavefunctions of the 2p subshell.

Q4. Level 1 • Spin and boundary surfaces

(a) Which values can m_s take, and how are they drawn? Which experiment revealed spin? (b) What is a boundary surface? What is the shape of the boundary surface of an s orbital, and of a p orbital?

Q5. Level 2 •• Energies and degeneracy in hydrogen

(a) Give the energy of the electron of hydrogen described by $\Psi_{3,2,-1}$ and by $\Psi_{3,0,0}$. What word describes this situation? (b) How many degenerate orbitals are there in the shell $n = 4$ of hydrogen?

Q6. Level 2 •• Ground or excited?

For the hydrogen atom, say which orbital is occupied, whether the atom is in its ground state or an excited state, and its energy, when the electron is described by (a) $\Psi_{1,0,0}$, (b) $\Psi_{2,1,0}$, (c) $\Psi_{3,2,1}$. (d) How much energy is needed to go from (a) to (b)?

Q7. Level 2 •• Counting nodes

Give the number of radial nodes, angular nodes and total nodes of the 2s, 3p, 3d, 4s and 4f orbitals. Check the 3s, 3p and 3d cases on Figure 3-3 of the handout.

Q8. Level 2 •• The three p orbitals

For each of p_x , p_y and p_z , give the axis along which it points and its nodal plane. Which one corresponds to $m_\ell = 0$? What happens to the sign of the wavefunction from one lobe to the other?

Q9. Level 3 ••• Size of orbitals

Using $r = n^2 a_0/Z$, compute the radius of: (a) the 1s, 2s and 3s orbitals of hydrogen; (b) the 1s orbital of He^+ and of Li^{2+} . (c) Explain in one sentence the effect of n and of Z .

Q10. Level 3 ••• Ordering subshells

Using Madelung's rule, write $n + \ell$ for each subshell and put them in order of increasing energy: 5s, 4f, 3d, 4p, 6s, 4d, 5p, 4s.

Q11. Level 3 ••• Hydrogen versus potassium

In hydrogen the 3d orbitals have the same energy as 3s, and lie far below 4s. In potassium ($Z = 19$) the 4s orbital is lower than 3d. (a) Which approximation lets us still use the labels 1s, 2s, 2p ... for many-electron atoms? (b) Explain why the energy now depends on ℓ and why 4s can drop below 3d.

Q12. Level 3 ●●● The Zeeman effect

- (a) Into how many sublevels does a p subshell split in a magnetic field? a d subshell? an f subshell?
(b) Hydrogen atoms emit the $2p \rightarrow 1s$ line (122 nm). How many lines are seen in a magnetic field?
(c) What does the Zeeman effect prove?

Q13. Challenge ★ How many electrons can a shell hold?

An electron is described by four quantum numbers (n, ℓ, m_ℓ, m_s) . (a) List all the allowed sets for $n = 2$. (b) Show that the number of sets for a shell n is $2n^2$ and check it for $n = 1, 2, 3$. (Unit 4: no two electrons of an atom can share the same set, so this is the maximum number of electrons of the shell.)

Q14. Challenge ★ The sodium doublet

The D line of sodium is a doublet: $\lambda_1 = 589.0$ nm and $\lambda_2 = 589.6$ nm. (a) Compute the energy of each photon in J and in eV (4 significant figures). (b) Compute the energy difference between the two upper levels in J and in meV. (c) Explain the origin of the doublet.

Q15. Challenge ★ Where is the 1s electron?

The radial probability density of the 1s orbital of hydrogen is $P(r) = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$. (a) Show that P is maximum at $r = a_0$. (b) Explain why the probability density is zero at $r = 0$ even though $|\Psi_{1s}|^2$ is largest at the nucleus. (c) For He^+ , $P(r) \propto r^2 e^{-2Zr/a_0}$: where is its maximum?

UNIT 4

Electronic configuration and periodic table

Handout chapter 4, sections 4.1–4.2 (pages 13–15). Three rules place the electrons of any atom in its orbitals; the resulting configuration explains the layout of the periodic table.

Concepts in detail

4.1 Electronic configuration in the ground state

Electronic configuration. The list of the occupied subshells with the number of electrons in each, written as a superscript: for example O: $1s^2 2s^2 2p^4$. The superscripts add up to the number of electrons (Z for a neutral atom). The ground-state configuration follows three rules.

4.1.1 Aufbau (building-up) principle

Aufbau principle. In the ground state, the electrons occupy the orbitals of lowest energy first. The order of the subshells is given by Madelung's rule (Unit 3): $1s\ 2s\ 2p\ 3s\ 3p\ 4s\ 3d\ 4p\ 5s\ 4d\ 5p\ 6s\ 4f\ 5d\ 6p\ 7s\ 5f\ 6d\ 7p$.

4.1.2 Pauli exclusion principle (1925)

Pauli exclusion principle. Two electrons of the same atom cannot have the same four quantum numbers (n, ℓ, m_ℓ, m_s). Consequence: one orbital (fixed n, ℓ, m_ℓ) holds **at most two electrons**, and they must have opposite spins ($\uparrow\downarrow$, “paired”).

Completing the handout: Table 4-1, maximum number of electrons in each subshell

Subshell	s ($\ell = 0$)	p ($\ell = 1$)	d ($\ell = 2$)	f ($\ell = 3$), for reference
Number of AOs ($2\ell + 1$)	1	3	5	7
Maximum number of electrons $2(2\ell + 1)$	2	6	10	14

A shell n holds at most $2n^2$ electrons: 2, 8, 18, 32.

4.1.3 Hund's rule

Hund's rule. In a subshell with several orbitals (p, d, f), the electrons first occupy *different* orbitals with *parallel* spins (same m_s) before pairing up. This configuration has the lowest energy: electrons in different orbitals repel each other less.

Completing the handout: box after Hund's rule, example of carbon and nitrogen

Each box is an orbital, each arrow an electron. Carbon, $1s^2 2s^2 2p^2$, and nitrogen, $1s^2 2s^2 2p^3$ (2p subshell only):

	correct (Hund)	wrong: paired too early	wrong: spins not parallel
C, $2p^2$	$\uparrow\uparrow\Box$	$\uparrow\downarrow\Box$	$\uparrow\downarrow\Box$
N, $2p^3$	$\uparrow\uparrow\uparrow$	$\uparrow\downarrow\uparrow$	$\uparrow\downarrow\uparrow$

Carbon has 2 unpaired electrons, nitrogen 3. (By convention the first electron of an orbital is drawn $\uparrow$.) The “wrong” diagrams are allowed by Pauli but describe *excited* states.

4.1.4 Mapping the first 18 elements

Filling in order $1s, 2s, 2p, 3s, 3p$ and applying Pauli and Hund gives:

Completing the handout: Table 4-2, electronic configurations from H ($Z = 1$) to Ar ($Z = 18$)

$n = 1$ (boxes: $1s$)					
H	$1s^1$	$\uparrow$	He	$1s^2$	$\uparrow\downarrow$
$n = 2$ (boxes: $2s$ $2p$; core $[\text{He}] = 1s^2$)					
Li	$[\text{He}] 2s^1$	$\uparrow$ $\square$ $\square$ $\square$	Be	$[\text{He}] 2s^2$	$\uparrow\downarrow$ $\square$ $\square$ $\square$
B	$[\text{He}] 2s^2 2p^1$	$\uparrow\downarrow$ $\uparrow$ $\square$ $\square$	C	$[\text{He}] 2s^2 2p^2$	$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\square$
N	$[\text{He}] 2s^2 2p^3$	$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$	O	$[\text{He}] 2s^2 2p^4$	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\square$
F	$[\text{He}] 2s^2 2p^5$	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\square$	Ne	$[\text{He}] 2s^2 2p^6$	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\square$
$n = 3$ (boxes: $3s$ $3p$; core $[\text{Ne}] = 1s^2 2s^2 2p^6$)					
Na	$[\text{Ne}] 3s^1$	$\uparrow$ $\square$ $\square$ $\square$	Mg	$[\text{Ne}] 3s^2$	$\uparrow\downarrow$ $\square$ $\square$ $\square$
Al	$[\text{Ne}] 3s^2 3p^1$	$\uparrow\downarrow$ $\uparrow$ $\square$ $\square$	Si	$[\text{Ne}] 3s^2 3p^2$	$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\square$
P	$[\text{Ne}] 3s^2 3p^3$	$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$	S	$[\text{Ne}] 3s^2 3p^4$	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\square$
Cl	$[\text{Ne}] 3s^2 3p^5$	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\square$	Ar	$[\text{Ne}] 3s^2 3p^6$	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\square$

The full form of Cl, for example, is $1s^2 2s^2 2p^6 3s^2 3p^5$. The shorthand writes the previous noble gas in brackets for the core.

Worked example: iron, $Z = 26$ Fill in Madelung order: $1s^2 2s^2 2p^6 3s^2 3p^6$ (18 electrons, $[\text{Ar}]$), then $4s^2$ (20), then 6 electrons in $3d$ (26): Fe: $[\text{Ar}] 4s^2 3d^6$, often written $[\text{Ar}] 3d^6 4s^2$ (by increasing n), as on the handout's periodic table. $3d$ boxes: $\uparrow\downarrow\uparrow\uparrow\uparrow\uparrow$: 4 unpaired electrons.

4.1.5 Core and valence electrons

Core electrons. Electrons in the low-energy AOs, close to the nucleus and strongly bound. They take no part in chemistry. They make up the configuration of the previous noble gas, written in brackets.

Valence electrons. Electrons in the highest-energy AOs, far from the nucleus and weakly bound: those of the highest n , plus those of an unfilled $(n - 1)d$ subshell for transition metals. They determine the chemical properties. They are the electrons written after the bracket.

Example: $\text{Cl} = [\text{Ne}] 3s^2 3p^5$ has 10 core electrons and 7 valence electrons. It needs one electron to reach the configuration of argon, which is why it forms Cl^- .

Configurations of ions. Anions: add electrons in the next free orbitals ($\text{Cl}^-: [\text{Ne}] 3s^2 3p^6 = [\text{Ar}]$). Cations: remove electrons from the *highest* n first. For transition metals the $4s$ electrons leave before the $3d$ ones: $\text{Fe}^{2+}: [\text{Ar}] 3d^6$, $\text{Fe}^{3+}: [\text{Ar}] 3d^5$ (not $[\text{Ar}] 4s^2 3d^3$).

4.2 The periodic table of the elements

4.2.1 Structure

- Elements are placed in order of increasing **atomic number** Z .
- **18 columns = groups**: elements of a group have the same valence configuration and similar chemistry (group 1: alkali metals, 2: alkaline-earth metals, 17: halogens, 18: noble gases).
- **7 rows = periods**: period n is the filling of the valence shell n (it starts with ns^1 and ends with np^6 , except period 1).
- **Blocks** named after the subshell being filled: s (groups 1–2, and He), p (13–18), d (3–12, transition metals), f (lanthanides and actinides, drawn below).
- **Mendeleev** (1869) ordered the known elements by atomic *mass* and grouped them by properties, leaving gaps for undiscovered elements (gallium, germanium) whose properties he predicted. Today the order is by Z , which also fixes the few inversions of mass (Ar/K, Co/Ni, Te/I).

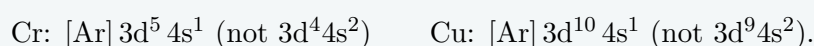
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1																		
2																		
3																		
4																		
5																		
6																		
7																		
	ns^{1-2}		$(n-1)d^{1-10} ns^2$										$ns^2 np^{1-6}$					
	lanthanides		f										(n-2)f					
	actinides																	

The length of the periods follows from Madelung's order: period 1 fills 1s (2 elements); periods 2 and 3 fill ns and np (8); periods 4 and 5 fill ns , $(n-1)d$, np (18); periods 6 and 7 also fill $(n-2)f$ (32).

4.2.2 Valence configuration from the position in the table

- Period number = largest n of the configuration.
- s and p blocks: number of valence electrons = the position in the period (group 1: 1, group 2: 2, groups 13–18: 3 to 8). Example: sulfur, period 3, group 16: 6 valence electrons, $[\text{Ne}] 3s^2 3p^4$.
- d block: valence configuration $(n-1)d^x ns^2$ with $x = \text{group} - 2$. Example: manganese, period 4, group 7: $[\text{Ar}] 3d^5 4s^2$.
- Elements of the same group have the same valence configuration with a different n : Li $2s^1$, Na $3s^1$, K $4s^1$...

Exceptions to Madelung's rule. The rule is always followed in the s and p blocks, but not always in the d and f blocks, where the ns and $(n-1)d$ levels are very close. A half-filled or filled d subshell is especially stable:



The same happens for Mo, Ag and Au below them.

Common mistake Count electrons, not protons, for an ion. When writing a cation of a transition metal, remove 4s before 3d. Do not confuse the group number (1–18) with the number of valence electrons in the d block: Fe is in group 8 and has 8 valence electrons ($3d^6 4s^2$), but in the p block chlorine is in group 17 and has 7.

Recap · Unit 4: electronic configuration and periodic table**The three filling rules**

Aufbau: lowest-energy orbitals first, in Madelung order 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s.

Pauli: no two electrons with the same (n, ℓ, m_ℓ, m_s) : at most 2 electrons per orbital, with opposite spins.

Hund: in a subshell, fill every orbital singly with parallel spins before pairing.

Capacities

s: 1 AO, 2 e^- ; p: 3 AOs, 6 e^- ; d: 5 AOs, 10 e^- ; f: 7 AOs, 14 e^- ; shell n : $2n^2 e^-$.

Method for a configuration

(1) Count the electrons (Z , minus the charge). (2) Fill in Madelung order. (3) Replace the core by the previous noble gas in brackets. (4) Draw boxes for the valence subshells using Pauli and Hund; count unpaired electrons. (5) Cations of d-block metals: remove ns first.

Periodic table

Order by Z ; 18 groups (same valence configuration, similar chemistry); 7 periods (period number = valence shell n); blocks s, p, d, f = subshell being filled. Period lengths 2, 8, 8, 18, 18, 32, 32. Valence configuration: s block ns^{1-2} , p block ns^2np^{1-6} , d block $(n-1)d^{1-10}ns^2$.

Key examples

O $[\text{He}]2s^22p^4$ (2 unpaired); Cl $[\text{Ne}]3s^23p^5$ (7 valence e^-); Fe $[\text{Ar}]3d^64s^2$; Fe^{3+} $[\text{Ar}]3d^5$; Cr $[\text{Ar}]3d^54s^1$; Cu $[\text{Ar}]3d^{10}4s^1$.

You should be able to

- write the ground-state configuration (full and shorthand) of any atom or ion up to $Z = 36$;
- draw box diagrams and count unpaired electrons;
- find the period, group and block from a configuration, and the reverse;
- separate core and valence electrons;
- recognise ground, excited and impossible configurations.

Worksheet 4 · Electronic configuration and periodic table

Data Atomic numbers: H 1, He 2, Li 3, Be 4, B 5, C 6, N 7, O 8, F 9, Ne 10, Na 11, Mg 12, Al 13, Si 14, P 15, S 16, Cl 17, Ar 18, K 19, Ca 20, Sc 21, Ti 22, V 23, Cr 24, Mn 25, Fe 26, Co 27, Ni 28, Cu 29, Zn 30, Ga 31, Ge 32, As 33, Se 34, Br 35, Kr 36, Rb 37, Sr 38. No calculator needed.

Q1. Level 1 • Capacity of subshells

Complete: an s subshell has ... orbital(s) and holds at most ... electrons; p: ... and ...; d: ... and ...; f: ... and Which principle limits an orbital to two electrons?

Q2. Level 1 • The three rules

State the Aufbau principle, the Pauli exclusion principle and Hund's rule in one sentence each.

Q3. Level 1 • Writing configurations

Write the full ground-state configuration of O, Na, Cl and Ar, then the shorthand form with a noble gas core.

Q4. Level 1 • Core and valence

Give the number of core electrons and of valence electrons of C, Mg, S and Ar.

Q5. Level 2 •• Box diagrams and Hund's rule

(a) Draw the box diagram of the valence shell of N, O and F and give the number of unpaired electrons. (b) Which of these $2p^3$ diagrams respects Hund's rule? Explain what is wrong with the others: (i) $\uparrow\downarrow\uparrow\uparrow$ (ii) $\uparrow\uparrow\uparrow\uparrow$ (iii) $\uparrow\uparrow\downarrow\uparrow$.

Q6. Level 2 •• From configuration to element

Identify the element and give its period, group and block: (a) $1s^2 2s^2 2p^6 3s^2 3p^4$; (b) $[\text{Ne}] 3s^1$; (c) $[\text{Ar}] 3d^6 4s^2$; (d) $[\text{Kr}] 5s^2$.

Q7. Level 2 •• From position to configuration

Write the shorthand configuration of the element located at: (a) period 3, group 15; (b) period 4, group 2; (c) period 4, group 17; (d) period 5, group 1. Name each element.

Q8. Level 2 •• Ground, excited or impossible?

Classify each configuration as ground state, excited state or impossible, and name the atom: (a) $1s^2 2s^2 2p^6 3s^1$; (b) $1s^2 2s^2 2p^5 3s^1$; (c) $1s^2 2s^3$; (d) $1s^2 2p^1$; (e) $1s^2 2s^2 2p^7$; (f) $1s^2 2s^2 2p^6 3d^1$.

Q9. Level 3 ••• Ions and noble gases

Write the configuration of Na^+ , Mg^{2+} , O^{2-} , Cl^- and S^{2-} . Which noble gas does each resemble? Explain why these ions are the ones usually formed.

Q10. Level 3 ••• Iron and its ions

(a) Write the configuration of Fe ($Z = 26$) and draw the box diagram of its 3d and 4s subshells. How many unpaired electrons does it have? (b) Write the configuration of Fe^{2+} and Fe^{3+} . (c) Fe^{3+} is particularly stable. Suggest why.

Q11. Level 3 ••• Exceptions: chromium and copper

(a) Write the configuration predicted by Madelung's rule for Cr ($Z = 24$) and Cu ($Z = 29$). (b) The real configurations are $[\text{Ar}] 3d^5 4s^1$ and $[\text{Ar}] 3d^{10} 4s^1$. Explain. (c) How many unpaired electrons do Cr and Cu have?

Q12. Level 3 ●●● **Four quantum numbers**

(a) Give a possible set (n, ℓ, m_ℓ, m_s) for each of the five valence electrons of nitrogen in its ground state. (b) Which of these sets could describe an electron of a ground-state phosphorus atom? (i) $(3, 1, 0, +\frac{1}{2})$; (ii) $(3, 2, 0, +\frac{1}{2})$; (iii) $(2, 1, -1, -\frac{1}{2})$; (iv) $(3, 0, 0, 0)$.

Q13. Challenge ★ **Why 2, 8, 8, 18, 18, 32?**

Using Madelung's order and the fact that each period starts when a new ns subshell starts filling, list the subshells filled in each of the first six periods and deduce the number of elements in each.

Q14. Challenge ★ **Magnetic atoms**

Atoms with unpaired electrons are attracted by a magnet (paramagnetic); atoms with none are not (diamagnetic). Find the number of unpaired electrons of Mn ($Z = 25$), Fe, Zn ($Z = 30$), Cu and Zn^{2+} . Which are diamagnetic? Which ion of manganese, Mn^{2+} or Mn^{3+} , has the most unpaired electrons?

Q15. Challenge ★ **Beyond the known table**

The last element of period 7 is oganesson, Og ($Z = 118$). (a) Write the valence configuration of Og. (b) Using Madelung's rule, which subshells would period 8 fill, and in which order? (Remember that $\ell = 4$ is called g.) (c) Predict the atomic number of the noble gas at the end of period 8.

UNIT 5

Atomic properties and periodicity

Handout chapter 5, sections 5.1–5.5 (pages 16–20). Five properties of atoms (mass, radius, ionisation energy, polarisability, electronegativity) and how they vary across the periodic table. Most trends are explained by two ideas: the shell n of the valence electrons and the effective nuclear charge they feel.

Concepts in detail

Key idea Effective nuclear charge, the key to every trend

Screening and effective nuclear charge Z_{eff} . A valence electron is attracted by the Z protons but repelled by the other electrons, especially the core electrons between it and the nucleus. It feels only an effective charge $Z_{\text{eff}} = Z - \sigma$, where σ (the screening constant) measures the shielding by the other electrons.

- **Across a period** (left to right): one proton and one valence electron are added each time, but electrons of the same shell screen each other poorly. Z_{eff} increases (roughly +0.65 per step), the valence shell n stays the same: valence electrons are held more tightly.
- **Down a group:** Z_{eff} changes little, but n increases by one: the valence electrons are in a larger shell, further from the nucleus, and held more loosely.

Going further (not required) Slater's rules estimate σ for an ns or np electron: each other electron of the same shell contributes 0.35 (0.30 in $1s$), each electron of shell $n - 1$ contributes 0.85, each electron of lower shells 1.00. Example, Cl ($[\text{Ne}] 3s^2 3p^5$): $\sigma = 6 \times 0.35 + 8 \times 0.85 + 2 \times 1.00 = 10.9$ (the 6 other valence electrons, then the 8 electrons of shell 2, then the 2 of shell 1), so $Z_{\text{eff}} = 17 - 10.9 = 6.1$. Na: $\sigma = 8 \times 0.85 + 2 = 8.8$, $Z_{\text{eff}} = 2.2$.

5.1 Atomic mass

Atomic mass. The mass of one atom, in atomic mass units (amu). It is close to the mass number A (protons + neutrons, each ≈ 1 amu); the electrons are negligible (Unit 1). Numerically, the atomic mass in amu equals the molar mass in g mol^{-1} .

Measurement: the mass spectrometer ionises the atoms, accelerates them and separates them by mass-to-charge ratio in a magnetic or electric field. It gives the mass of each isotope and its abundance.

Completing the handout: bullets of 5.1.2, trend in the periodic table

- The atomic mass increases with Z along a period: each step adds a proton and usually one or more neutrons.
- It increases down a group (a whole period of nucleons is added). There are a few *inversions*, where mass and Z are not in the same order: Ar/K, Co/Ni, Te/I.

Average atomic mass. Most elements are mixtures of isotopes. The tabulated atomic mass is the average of the isotope masses weighted by their natural abundances x_i (fractions, $\sum x_i = 1$):

$$\bar{m} = \sum_i x_i m_i.$$

Completing the handout: chlorine, argon and potassium example

- Cl: $\bar{m} = 0.7577 \times 34.97 + 0.2423 \times 36.97 = 26.50 + 8.96 = \mathbf{35.45}$ amu (closer to 35 because ^{35}Cl is more abundant).
- Ar: almost pure ^{40}Ar : $\bar{m} \approx 39.96$ amu (the 0.4 % of ^{36}Ar and ^{38}Ar lowers it to 39.95).
- K: $\bar{m} = 0.9326 \times 38.96 + 0.0673 \times 40.96 = 36.33 + 2.76 = \mathbf{39.09}$ amu.

Conclusion: argon ($Z = 18$) is heavier than potassium ($Z = 19$). Ordering by mass, as Mendeleev did, would put K before Ar, in the column of the noble gases, which is chemically absurd. The periodic table is ordered by Z , not by mass.

5.2 Atomic radius

Atomic radius. The distance from the nucleus to the outer edge of the electron cloud: the radius of the highest occupied AO, i.e. the position of the maximum of its radial probability density. Since the density fades out gradually, it is measured indirectly from distances between nuclei in molecules and solids.

Covalent radius	half the distance between two identical bonded atoms (H_2 , Cl_2 ...)
Metallic radius	half the distance between neighbouring atoms in a metal crystal (Cu, Au)
Van der Waals radius	half the distance between two identical non-bonded atoms touching in a solid; much larger than the covalent radius (used for noble gases, marked “vdw” in Table 5-1)
Ionic radius	from distances between ions in an ionic crystal; cation < atom < anion

Completing the handout: bullets and box of 5.2.2, trend of the atomic radius

- The radius **increases down a group** (Li 157, Na 191, K 235, Rb 250, Cs 272 pm).
- The radius **decreases across a period** (Li 157 $\rightarrow$ F 64 pm; Na 191 $\rightarrow$ Cl 99 pm). The largest atoms are at the bottom left (Cs), the smallest at the top right.

Down a group: each period adds a shell; the valence electrons are in an orbital of larger n ($r \propto n^2/Z_{\text{eff}}$), while Z_{eff} barely changes. The atom gets bigger.

Across a period: same valence shell n , but Z_{eff} increases (poor screening within a shell), so the cloud is pulled closer to the nucleus. The atom gets smaller.

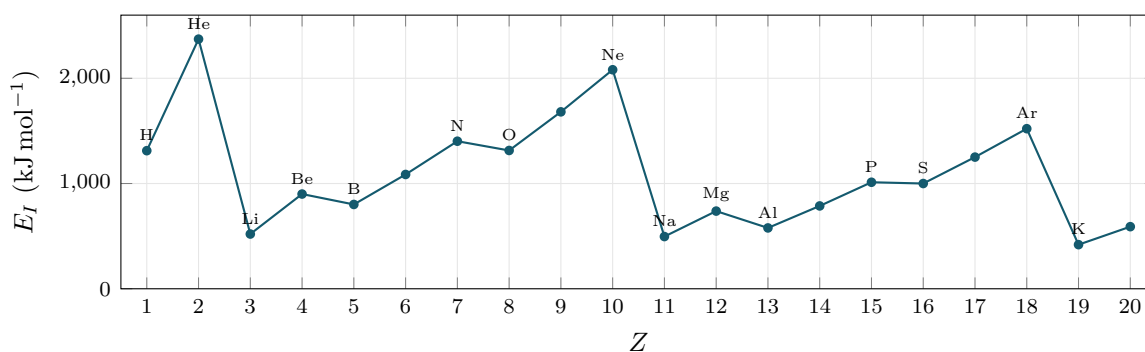
The noble gas values (“vdw”) are van der Waals radii and cannot be compared with the others: that is why Ne (154 pm) looks larger than F (64 pm).

Ions. A cation is smaller than its atom (fewer electrons, less repulsion; often a whole shell lost: Na^+ is [Ne]). An anion is larger (extra electron, more repulsion, same nuclear charge). In an isoelectronic series the radius decreases as Z increases: $r(\text{O}^{2-}) > r(\text{F}^-) > r(\text{Na}^+) > r(\text{Mg}^{2+})$ (all 10 electrons, 8 to 12 protons).

5.3 Ionisation energy

(First) ionisation energy E_I . The minimum energy needed to remove the outermost electron from an atom in the gas phase: $X(g) \rightarrow X^+(g) + e^-$. Given in kJ mol^{-1} in chemistry (1 eV per atom = 96.5 kJ mol^{-1}). It is always positive. The second ionisation energy removes a second electron from X^+ , and so on; each is larger than the previous one.

Measurement: by atomic spectroscopy, from the series limit (for H, the Lyman limit at 91.2 nm gives 13.6 eV, Unit 2), or by photoelectron spectroscopy (PES): a photon of known energy $h\nu$ ejects an electron whose kinetic energy E_k is measured, and $E_I = h\nu - E_k$.



First ionisation energies of the first 20 elements (Table 5-2; H: 1312).

Completing the handout: bullets and box of 5.3.2, trend of the ionisation energy

Hydrogen, missing in Table 5-2: $E_I = 13.6 \text{ eV} = 1312 \text{ kJ mol}^{-1}$.

- E_I **increases across a period** (Li 520 $\rightarrow$ Ne 2081): Z_{eff} increases and the atom shrinks, so the electron is held more strongly. Noble gases have the highest values.
- E_I **decreases down a group** (He 2372 $\rightarrow$ Rn 1037; Li 520 $\rightarrow$ Cs 376): the electron is further from the nucleus. Alkali metals have the lowest values.
- The increase across a period has two small **dips**: group 2 $\rightarrow$ 13 (Be 900 > B 801, Mg 738 > Al 578) and group 15 $\rightarrow$ 16 (N 1402 > O 1314, P 1012 > S 1000).

Be > B: the electron removed from B is its single 2p electron, which is higher in energy (and screened by 2s) than the 2s electron removed from Be. Be $\uparrow\downarrow \square\square\square$ vs B $\uparrow\downarrow \uparrow\square\square$.

N > O: N has a half-filled 2p ($\uparrow\uparrow\uparrow\uparrow$), each electron alone in its orbital. In O ($\uparrow\downarrow\uparrow\uparrow$) one orbital holds a pair; their mutual repulsion makes one of them easier to remove.

Successive ionisation energies. They increase slowly while valence electrons are removed, then jump enormously when the first core electron is reached. The position of the jump gives the number of valence electrons. Mg: 738, 1451, **7733**, 10 543 kJ mol^{-1} : the jump after the 2nd electron shows 2 valence electrons (group 2).

5.4 Polarisability

Polar entity and dipole moment. An entity is polar when the barycentres of its positive and negative charges do not coincide. It is modelled by two charges $+q$ and $-q$ at a distance d , with dipole moment $\vec{\mu} = q\vec{d}$, from $-q$ to $+q$ (the chemists' convention). SI unit: C m; in chemistry the debye, $1 \text{ D} = 3.336 \times 10^{-30} \text{ C m}$.

Polarisability α . How easily the electron cloud of an atom is distorted by an external electric field $\vec{E}$, creating an induced dipole $\vec{\mu}_{\text{ind}} = \alpha\vec{E}$. A large, diffuse, loosely held cloud is very polarisable.

Measurement: from the change of the refractive index of the substance in an electric field (light itself is an oscillating field that distorts the clouds).

Completing the handout: bullets of 5.4.2, trend of the polarisability

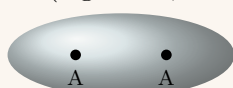
- Polarisability **increases down a group**: larger atoms, valence electrons further from the nucleus and better screened ($I > Br > Cl > F$; $Xe > Kr > Ar$).
- It **decreases across a period**: the atoms shrink and Z_{eff} holds the cloud more tightly. It follows the atomic radius. Anions are more polarisable than cations.

Consequence: large polarisable atoms attract each other more (London forces), which is why the boiling points rise from F_2 (gas) to I_2 (solid) and from He to Rn.

5.5 Electronegativity

Completing the handout: box of 5.5, bond A–A versus bond A–B

A–A (e.g. H–H, Cl–Cl)



The shared electrons are equally attracted by both atoms: symmetric density, **non-polar** bond.

A–B (e.g. H–Cl)



B attracts the bonding electrons more than A: the density is shifted towards B, which carries a partial charge δ^- (A: δ^+). The bond is **polar** and has a dipole moment.

Electronegativity χ . The ability of an atom, *in a bond*, to attract the shared electron density towards itself. It is a relative number without unit. The larger $\chi_B - \chi_A$, the more polar the bond A–B; for a very large difference ($\gtrsim 2$) the bond becomes ionic (NaCl).

5.5.1 Measurement: two scales

Pauling (experimental). A polar bond A–B is *stronger* than expected from the bonds A–A and B–B, because the partial charges attract. The excess energy $\Delta = D_{AB} - \sqrt{D_{AA} D_{BB}}$ (bond dissociation energies in kJ mol^{-1}) gives

$$|\chi_A - \chi_B| = 0.102 \sqrt{\Delta}$$

and the scale is anchored by $\chi_H = 2.20$ (or $\chi_F = 3.98$).

Worked example: H–F from Table 5-3 $\sqrt{436 \times 158} = 262.5$; $\Delta = 567 - 262.5 = 304.5 \text{ kJ mol}^{-1}$; $|\chi_F - \chi_H| = 0.102\sqrt{304.5} = 1.78$, so $\chi_F = 2.20 + 1.78 = 3.98$, the value of Table 5-4. For H–I the same formula gives 0.66 (so $\chi_I \approx 2.86$ instead of 2.66): the tabulated values are a best fit over many molecules, and a single pair only gives an estimate. (Pauling's original version used the arithmetic mean $\frac{1}{2}(D_{AA} + D_{BB})$ and energies in eV: $|\chi_A - \chi_B| = \sqrt{\Delta/(96.5 \text{ kJ mol}^{-1})}$; use the version given in class.)

Allred–Rochow (theoretical). Based on the electrostatic force felt by a valence electron at the covalent radius r : $\chi = 0.359 Z_{\text{eff}}/r^2 + 0.744$ (r in Å). Large Z_{eff} and small radius mean high electronegativity. Both scales give very similar values.

Completing the handout: bullets of 5.5.2, trend of the electronegativity

- χ **increases across a period** (Li 0.98 $\rightarrow$ F 3.98): Z_{eff} increases and the radius decreases. **Fluorine** is the most electronegative element, then O (3.44), then N and Cl.
- χ **decreases down a group** (F 3.98 $\rightarrow$ I 2.66; Li 0.98 $\rightarrow$ Cs 0.79): larger atoms hold bonding electrons less. Noble gases have no value (they hardly form bonds).

Metals (bottom left) have low χ and form cations; non-metals (top right) have high χ .

Summary	across a period $\rightarrow$ (Z_{eff} $\nearrow$, same n)	down a group $\downarrow$ (n $\nearrow$)
atomic radius	decreases	increases
ionisation energy	increases	decreases
polarisability	decreases	increases
electronegativity	increases	decreases
atomic mass	increases	increases

Common mistake Do not mix up electronegativity (an atom *in a bond* attracting shared electrons) and ionisation energy (an *isolated* atom losing an electron). Compare radii only within the same type (covalent with covalent). “Across a period” trends apply to the s and p blocks; in the d block the changes are much smaller.

Recap · Unit 5: atomic properties and periodicity

Definitions

Atomic mass: mass of one atom in amu; tabulated value = abundance-weighted average of the isotopes, $\bar{m} = \sum x_i m_i$.

Atomic radius: radius of the highest occupied AO; measured as covalent, metallic, van der Waals or ionic radius.

Ionisation energy: minimum energy for $X(g) \rightarrow X^+(g) + e^-$, in kJ mol^{-1} .

Polarisability: ease of distortion of the electron cloud by an electric field; dipole $\mu = qd$ (1 D = 3.336×10^{-30} C m).

Electronegativity χ : ability of an atom in a bond to attract the shared electrons; no unit.

Trends

	across a period $\rightarrow$	down a group $\downarrow$	extremes
radius	$\searrow$	$\nearrow$	Cs largest, He/F smallest
ionisation energy	$\nearrow$ (dips at 13 and 16)	$\searrow$	He highest, Cs lowest
polarisability	$\searrow$	$\nearrow$	follows the radius
electronegativity	$\nearrow$	$\searrow$	F 3.98 highest, Cs/Fr lowest

Explanation: across a period Z_{eff} increases with the same n ; down a group n increases.

Formulas and numbers

$\bar{m} = \sum x_i m_i$ (Cl 35.45, K 39.09, Ar 39.95). 1 eV/atom = 96.5 kJ mol^{-1} . PES: $E_I = h\nu - E_k$.

Pauling: $|\chi_A - \chi_B| = 0.102\sqrt{\Delta}$, $\Delta = D_{AB} - \sqrt{D_{AA}D_{BB}}$ (kJ mol^{-1}). Allred–Rochow: $\chi = 0.359 Z_{\text{eff}}/r^2 + 0.744$ (r in Å).

You should be able to

- compute an average atomic mass, or an abundance from an average mass;
- rank atoms and isoelectronic ions by radius, ionisation energy, polarisability or electronegativity, and justify with n and Z_{eff} ;
- explain the Be/B and N/O anomalies with box diagrams;
- read successive ionisation energies to find the group;
- use the Pauling formula and judge the polarity of a bond.

Worksheet 5 · Atomic properties and periodicity

Data $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $c = 3.00 \times 10^8 \text{ m/s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $N_A = 6.022 \times 10^{23} / \text{mol}$
 $1 \text{ D} = 3.336 \times 10^{-30} \text{ C} \cdot \text{m}$. Use Tables 5-1 to 5-4 of the handout (radii, ionisation energies, bond energies, Pauling electronegativities). Calculator allowed.

Q1. Level 1 • Definitions

Define: atomic radius, first ionisation energy, polarisability, electronegativity. For each, say whether it is a property of an isolated atom or of an atom in a bond.

Q2. Level 1 • Average atomic mass of chlorine

Chlorine is 75.77% ^{35}Cl (34.97 amu) and 24.23% ^{37}Cl (36.97 amu). Compute its average atomic mass and its molar mass.

Q3. Level 1 • Ranking radii

Using Table 5-1, rank Na, Mg, K and Cl by increasing atomic radius and state the two trends that explain the order.

Q4. Level 1 • Units of ionisation energy

The ionisation energy of sodium is 496 kJ mol^{-1} . Convert it to joules per atom and to electronvolts per atom. Show that 1 eV per atom corresponds to 96.4 kJ mol^{-1} .

Q5. Level 2 •• Mass versus atomic number

- (a) Compute the average atomic mass of potassium: ^{39}K 38.96 amu, 93.26%; ^{41}K 40.96 amu, 6.73%.
 (b) Argon (39.95 amu) has $Z = 18$ and potassium $Z = 19$. What problem would appear if elements were ordered by mass? How is it solved today?

Q6. Level 2 •• Explaining the radius trends

Explain, using the valence shell n and the effective nuclear charge, why the radius decreases from Li (157 pm) to F (64 pm) and increases from F to I (133 pm). Why can Ne (154 pm) not be compared with F?

Q7. Level 2 •• An isoelectronic series

Rank O^{2-} , F^- , Na^+ and Mg^{2+} by increasing radius. Justify. Compare also the radius of Na^+ with that of Na, and of Cl^- with that of Cl.

Q8. Level 2 •• The ionisation energies of period 2

The values for period 2 are: Li 520, Be 900, B 801, C 1086, N 1402, O 1314, F 1681, Ne 2081 kJ mol^{-1} .
 (a) State the general trend and explain it. (b) Explain the two exceptions with box diagrams.

Q9. Level 3 ••• Ionising light

- (a) Compute the longest wavelength of a photon able to ionise a sodium atom (496 kJ mol^{-1}), and the one for helium (2372 kJ mol^{-1}). In which region of the spectrum are they? (b) In a PES experiment, UV photons of 21.2 eV hit sodium atoms. What is the kinetic energy of the ejected valence electrons?

Q10. Level 3 ••• Successive ionisation energies

An element of period 3 has successive ionisation energies 738, 1451, 7733, $10\,543 \text{ kJ mol}^{-1}$. (a) Identify it and justify. (b) Why does each ionisation energy exceed the previous one? (c) Which ion does this element form in its compounds?

Q11. Level 3 ●●● Abundance from the average mass

Boron has two isotopes, ^{10}B (10.013 amu) and ^{11}B (11.009 amu), and an average atomic mass of 10.81 amu. Compute the natural abundance of each isotope.

Q12. Level 3 ●●● How polar is H–Cl?

The dipole moment of HCl is 1.08 D and its bond length 127 pm. (a) Compute the partial charge δ (as a fraction of e) carried by each atom. Which atom is δ^- ? (b) Using Table 5-4, rank H–F, H–Cl, H–Br and H–I by increasing polarity.

Q13. Challenge ★ Pauling electronegativity from bond energies

Use Table 5-3 (H–H 436, F–F 158, H–F 567, I–I 151, H–I 299 kJ mol $^{-1}$), the formula $|\chi_A - \chi_B| = 0.102\sqrt{\Delta}$ with $\Delta = D_{AB} - \sqrt{D_{AA}D_{BB}}$, and $\chi_{\text{H}} = 2.20$. (a) Compute χ_{F} . (b) Compute χ_{I} and compare with Table 5-4 (2.66). (c) Why is the H–F bond much stronger than the average of H–H and F–F?

Q14. Challenge ★ The Allred–Rochow scale

(a) Using Slater's rules (Unit 5, "Going further"), show that $Z_{\text{eff}} = 6.1$ for a valence electron of Cl. (b) With $\chi = 0.359 Z_{\text{eff}}/r^2 + 0.744$ (r in Å) and the covalent radius of Cl (99 pm), compute $\chi_{\text{AR}}(\text{Cl})$ and compare with Pauling's 3.16. (c) Without calculation, explain why F has a higher electronegativity than Cl on this scale.

Q15. Challenge ★ Synthesis: an unknown element

Element X is in period 3. Its successive ionisation energies are 578, 1817, 2745 and 11 577 kJ mol $^{-1}$. (a) Identify X and write its configuration with the box diagram of its valence shell. (b) Explain why the first ionisation energy of X is lower than that of Mg (738), although X has one more proton. (c) Compare the radius, the electronegativity and the polarisability of X with those of Mg and Si, using the tables. (d) Which ion does X form? Is it isoelectronic with a noble gas?